

Rigid Analytic Geometry Week 4 Solution

So Murata

SoSe 26, University of Bonn

Problem 1

Suppose $x \in L^{\circ m}$. Then for any $f \in \mathbb{T}_m$, we have,

$$|f(x)| \leq \max\{|f_\alpha x^\alpha| \mid \alpha \in \mathbb{N}^m\} \leq \max\{|f_\alpha| \mid \alpha \in \mathbb{N}^m\} = \|f\|_{\mathbb{T}_m}.$$

Problem 2

The implication from the third to the first is clear.

Suppose $f \in \mathbb{T}_m$ is not invertible. Then there is a maximal ideal containing it. By proposition, $f \in \mathfrak{m}_x$ for some $x \in L$ where L/K is a finite extension. Thus $f(x) = 0$. Taking the contrapositive, we have the implication.

Clearly, if $f \in \mathbb{T}_m^\times$, then it is nowhere 0. If $f \in \mathbb{T}_m^\times$. Observe that $|f^{-1}(x)| = |f(x)|^{-1}$, $\|f^{-1}\|_{\mathbb{T}_m} = \|f\|_{\mathbb{T}_m}^{-1}$. Using the inequality,

$$|f(x)| \leq \|f\|_{\mathbb{T}_m}, |f(x)^{-1}| \leq \|f^{-1}\|_{\mathbb{T}_m} \iff |f(x)| \geq \|f\|_{\mathbb{T}_m},$$

we get the equality.

Problem 3

Fix $x_0 \in K^\circ$. By setting $f \mapsto f(\cdot + x_0)$, we assume $x_0 = 0$. Similarly, by setting $f \mapsto f(\frac{\cdot}{R})$, we reduce to the case where $R = 1$. Then by Problem 2, $|f(y)| = \|f\|_{\mathbb{T}_m}$.

Problem 4

Let $(x_i)_{i \in \mathbb{N}}$ be the sequence in L approximating $x \in L$. As L is perfect, there is $(y_i)_{i \in \mathbb{N}}$ such that $y_i^p = x_i$ for the characteristic p of L . Note that

$$y_i^p - y_j^p = (y_i - y_j) \left(\sum_{k=0}^{p-1} y_i^k y_j^{p-1-k} \right) = x_i - x_j.$$

Note also that $(|y_i|)_{i \in \mathbb{N}}$ is bounded. With this, we conclude $(|y_i - y_j|)_{i \in \mathbb{N}}$ is Cauchy thus admits a limit y in $\mathbb{L}$. Since $(\cdot) \mapsto (\cdot)^p$ is continuous, $y^p = x$.

Problem 5

Let $G = \text{Gal}(L/K)$. We first show that $\sigma \in G$ preserves metric. Indeed, let us define,

$$\|\cdot\|_\sigma = |\sigma(\cdot)|_L.$$

Since σ fixes K , this defines a K -norm on L . Thus by the uniqueness of $|\cdot|_L$, we have,

$$|\sigma(x)|_L = |x|_L$$

for any $x \in L$. Since $K(\beta)$ is a subfield of L , there is $H \subseteq G$ such that $L^H = K(\beta)$. Let $\sigma \in H$ then,

$$|\beta - \alpha| = |\sigma(\beta - \alpha)| = |\beta - \sigma(\alpha)|.$$

Under the assumption, we get $\sigma(\alpha) = \alpha$. In particular $\alpha \in L^H = K(\beta)$.

Problem 6

Let $R = \max\{|a_1|, \dots, |a_n|\}$. Since coefficients p_i are elementary symmetric polynomials in a_j , we have,

$$|p_i| = \left| \sum_{\alpha \in \mathbb{N}^d, |\alpha|=d-i} a^\alpha \right| \leq R^{d-i}.$$

Indeed, the equality holds as a_j is in the summands where $|a_j| = R$. That is,

$$|a_j^{d-i}| \leq \max\{|p_i|, |p_i - a_j^{d-i}|\} \leq |a_j^{d-i}|.$$

We then conclude the statement.